Does a move he would find hold up as well as the engine's best?

Maia-2 plays on from the player's turning points and plan checkpoints

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Abstract

Game Review's better move is always the engine's top line. A human coach often shows another move instead: one the player would find, nearly as good. In the player's 10-minute games, at 73% of turning points and plan checkpoints the sound move players at his band would most likely play is already the engine's best. In the other 543 positions, each move was played on 16 times by Maia-2 at both players' bands for 12 plies, and the ends were scored by Stockfish. Both controls held. Against the engine's best, the move he would find scored −0.93 (−2.18 to 0.28) win% on the train games and −1.88 (−3.37 to −0.46) on the held-out games, so by the rule fixed before the run the verdict is *unclear*: a cost of more than one win% is neither ruled out nor shown. Over both sets it cost 1.27 (0.34 to 2.21) win% in human hands, against the 1.01 (0.94 to 1.08) the engine's own evaluation says it gives up: no hidden practical edge, and no hidden trap. The cost is clearest when the engine's best is forcing and may be nothing when it is quiet. At his turning points the move he would find scores 14.64 (12.81 to 16.44) win% above the move he played.

1 The question

Game Review names the engine's top line as the better move. A coach who knows the player often shows a different one: a move he would find, close to the best, with a follow-up he can play. The review's own window already treats a move within 3 win% of the best as sound, so such a move is never called an error. What the window cannot say is how the move fares when people at his level play on from it. A move the engine rates a little lower may be easier to handle, and score better in practice; or the engine's best may hold something the natural move gives away, which shows only later.

In each position, three moves:

- **B**, the engine's best;
- **F**, the move he would find: the sound move Maia-2 rates most likely at his band;
- **P**, the move he played.

The question is whether F, played on by people at these bands, scores as well as B.

2 How it was measured

The positions. His 10-minute games (no increment) in the app's archive at commit `f4bd63f19b`: 647 games, in order of their end time, the earlier two thirds (431) for training and the later third (216) held out. Nothing is fitted, so the two sets are two replications. Two kinds of position, both with him to move: the review's turning points, with up to five engine lines from the routine's deep

search, and its plan checkpoints, with up to three. The candidates are the first moves of those lines, each with its stored evaluation; a position needs two. A candidate is *sound* when it gives up no more than 3 win% against B, the review's own window. His band is the Maia-2 level of his rating at the game with the app's offset (+250), and the opponent's band the same of the opponent's rating. F is the sound candidate Maia-2 rates most likely at his band, conditioned on the opponent's band.

Of the 2,013 positions examined, 2,011 have two candidates or more, and in 1,468 of those (73%) F is B: the move his band would most likely play among the sound ones is the engine's best. The other 543 enter the study: 363 turning points and 180 plan checkpoints, 352 in the train games and 191 held out.

The rollouts. Each entering position plays out B and F, and P at a turning point when it is neither. Each move gets 16 rollouts. A rollout plays the move, then 12 more plies (six by each side), each sampled from Maia-2's policy for the side to move: his band for his moves, the opponent's band for the opponent's, each conditioned on the other's. Sampling is seeded by the position and the rollout. A rollout stops early when the game ends. Its end is scored as his win% by Stockfish through the app's engine service at the check budget (120,000 nodes); a mate scores 100 or 0 and a draw 50. 22,784 rollouts in all.

$V(m)$ is the mean of a move's 16 rollouts, and the measures are paired by position: $D_F = V(F) - V(B)$, what the move he would find costs or gains in the hands of players at these bands, and $D_P = V(P) - V(B)$, the same for his move. Intervals are the 95% percentiles of 2,000 bootstrap draws over games.

The rule, fixed before any rollout was scored. The controls come first, and if either fails on either set there is no verdict. *The null*: B's rollouts split into the first 8 and the last 8, and the mean difference's interval must cover zero. *The decoy*: at the turning points where his move lost at least 10 win% and is neither B nor F, the interval of the mean D_P must lie below zero; rollouts that cannot see his known errors cannot judge F either. Then the verdict on F, from the mean D_F on each set: it *holds up* when the interval's lower end is above −1 win% on both sets (one win% is a third of the review's window); it is *better in his hands* when the lower end is above zero on both; *the engine's best is worth the trouble* when the upper end is below −1 on both; and it is *unclear* otherwise.

3 The controls and the verdict

	Train	Held out
Games	228	135
Positions	352	191
Null: B's two halves	−0.32 (−1.62 to 1.02)	0.63 (−1.00 to 2.31)
Decoy: his errors against B, D_P	−15.80 (−18.20 to −13.37)	−18.71 (−21.82 to −15.67)
$D_F = V(F) - V(B)$	**−0.93 (−2.18 to 0.28)**	**−1.88 (−3.37 to −0.46)**

Table 1: The controls and the verdict's measure, in win% with 95% intervals from 2,000 bootstrap draws over games. The decoy rows are the turning points where his move lost at least 10 win% and is neither B nor F.

Both controls held on both sets (Table 1). Two halves of the same move's rollouts agree within noise, and the rollouts see his errors clearly: at his turning points where he lost at least 10 win% (189 train, 113 held out), his move scores far below B in human hands too.

The move he would find scored a little below the engine's best on both sets. On the train games the interval reaches above zero; on the held-out games it lies below zero but not below -1. Neither set rules out a cost of more than one win%, and neither shows one, so the verdict is *unclear*.

4 What the rollouts describe

	Positions	Games	D_F, win%
All	543	363	−1.27 (−2.21 to −0.34)
B quiet	448	321	−0.85 (−1.85 to 0.16)
B forcing (a capture or a check)	95	83	−3.24 (−6.17 to −0.44)
Turning points	363	280	−1.55 (−2.86 to −0.29)
Plan checkpoints	180	164	−0.68 (−2.05 to 0.66)
Opening	133	124	−0.10 (−1.93 to 1.63)
Middlegame	272	210	−1.57 (−2.93 to −0.21)
Endgame	138	104	−1.78 (−3.92 to 0.11)

Table 2: D_F over both sets, by the kind of the engine's best, by the kind of position and by the divider's phase. These splits were named before the run but carry no rule: they describe.

These numbers describe; none of them carries a rule.

The cost is about what the engine says. Over both sets the move he would find gives up 1.27 (0.34 to 2.21) win% in human hands. The engine's own evaluation of the same moves says it gives up 1.01 (0.94 to 1.08). The interval of the first covers the second: in the hands of players at these bands, the engine's gap neither closes nor opens. F scored above B in 44% of positions, as a cost near zero with noisy rollouts would give.

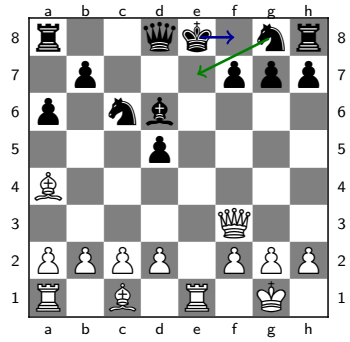
F is the move people play. At his band Maia-2 plays B 9% of the time and F 33% of the time, on average over the entering positions.

The cost sits with forcing moves. When B is quiet (448 positions) the cost's interval reaches zero. When B is a capture or a check (95), F costs more, and the interval lies below zero. The cost is larger at turning points than at plan checkpoints, and larger in the middlegame and endgame than in the opening (Table 2).

What costs him points is elsewhere. At the 338 turning points where he played neither B nor F, F scores 14.64 (12.81 to 16.44) win% above the move he played. Choosing the natural sound move over the engine's best costs about a point; missing both costs many.

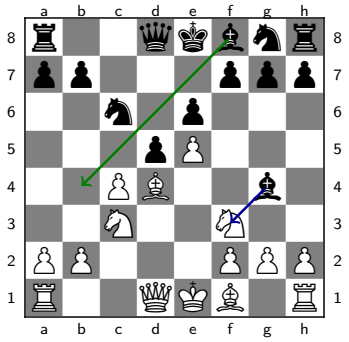
5 Two positions

One position's rollouts are noisy: two halves of the same move's rollouts differ by a median of 7.8 win%, and by 21.0 or more in one position in ten. The two positions below illustrate the summary; they do not test anything.



FEN `r2qk1nr/1p3ppp/p1nb4/3p4/B7/5Q2/PPPP1PPP/R1B1R1K1 b kq - 1 11`

(a) Black to move, a plan checkpoint in the opening, in check. The engine's best (blue) is 11. . . ♔f8 (`e8f8`), which his band plays 3% of the time; the move he would find (green) blocks, 11. . . ♘e7 (`g8e7`), at 79%. The engine sees a gap of 2.0 win%; in the rollouts F scored 76.1 and B 51.3, a difference of 24.9. He played 11. . . ♘e7.



FEN `r2qkbnr/pp3ppp/2n1p3/3pP3/2PB2b1/2N2N2/PP3PPP/R2QKB1R b KQkq - 0 8`

(b) Black to move, a turning point in the opening. The engine's best (blue) is the capture 8. . . ♗xf3 (`g4f3`), which his band plays 12% of the time; the move he would find (green) is 8. . . ♗b4 (`f8b4`), at 29%. The engine sees a gap of 1.9 win%; in the rollouts B scored 55.6 and F 42.0, a difference of −13.6. He played 8. . . ♘xd4 (`c6d4`).

Figure 1: Left: of the positions where both moves are quiet, the engine sees a gap of at least 1.5 win% and F scored at least as well as B, the one where Maia-2 rates F most above B. Right: of the positions where B is forcing, F is quiet and the engine's gap is at most 2 win%, the one where F scored furthest below B.

6 What the app can take

- The review can show the move he would find beside the engine's best when the best is quiet, as what it is: a sound move he would find, costing about what the review's window says. The study does not show it is better.
- When the engine's best is a capture or a check, the review should keep pointing to it: there the move he would find costs more in human hands.
- The larger gain is in the moves he plays that are neither. Coaching him toward a sound move he would find, whichever it is, is worth far more than coaching him from that move to the engine's.

7 Caveats

- **Maia-2 plays both sides.** The rollouts model players at these bands, not him or this opponent. A model of him is a later study.

- **12 plies.** The end is scored by Stockfish, so what comes after counts only as far as the engine sees it then.
- **Only the stored lines.** F is the most natural of the two to five moves the review kept, not of every legal move.
- **Near a decided game the window is wide.** Win% flattens close to a win or a loss, so moves the window admits can differ by more than their win% shows; the rollouts see it, the window does not.
- **The splits describe.** A second round could test quiet best moves alone, on positions this round did not see, with its margin fixed first.